Pharmaceutical Reform and Pharmacy Entry

County-level Regression Results, v5

Wenxia Chen

June 17, 2026

County-level Empirical Setting

- Unit of analysis: county-by-year panel, 2012–2022.
- Main controlled sample: 2012–2019, excluding counties treated in 2012, with population, nighttime lights, and road-density controls.
- Main treatment indicator:

$$DID_{ct} = \mathbf{1}\{t \geq PolicyYear_c\}.$$

- Baseline TWFE specification:

$$Y_{ct} = \beta DID_{ct} + X'_{ct}\gamma + \alpha_c + \lambda_t + \varepsilon_{ct}.$$

- Preferred high-dimensional FE specification replaces year FE with province-by-year FE:

$$Y_{ct} = \beta DID_{ct} + X'_{ct}\gamma + \alpha_c + \delta_{p(c)t} + \varepsilon_{ct}.$$

- Standard errors are clustered at the county level.

Controls include log population, log nighttime lights, and log road density where available.

County Sample and Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75
Pharmacy count	17,004	95.22	108.77	26	63	128
Log pharmacy count	15,512	4.14	1.15	3.54	4.26	4.92
Has pharmacy	17,004	0.91	0.28	1	1	1
Chain pharmacy count	17,004	8.09	27.30	0	0	1
Non-chain pharmacy count	17,004	87.13	96.55	25	59	117
DID	17,004	0.64	0.48	0	1	1
Log population	17,004	12.90	0.78	12.48	12.97	13.47
Log nighttime lights	17,004	0.72	0.92	0.08	0.27	1.03
Log road density	17,004	0.88	0.57	0.47	0.74	1.16

Source: county-level descriptive statistics from `desc_county.rtf`. Main sample uses observations with road-density controls.

Sample Support by Year

Year	County-year N	Road sample share	No-road sample share	DID share
2012	2,079	0.770	0.770	0.030
2013	2,194	0.773	0.773	0.045
2014	2,313	0.775	0.775	0.226
2015	2,623	0.780	0.780	0.556
2016	2,974	0.786	0.786	0.658
2017	3,135	0.788	0.788	1.000
2018	3,242	0.787	0.787	1.000
2019	3,176	0.789	0.789	1.000
2020	3,175	0.000	0.805	1.000
2021	3,184	0.000	0.000	1.000
2022	3,181	0.000	0.000	1.000

The main controlled specifications use the road sample through 2019. The no-road robustness column extends to 2020.

TWFE DID: Pharmacy Count

	FE	+ controls	+ prov-year	2012–2020 no road
Post policy	7.652***	5.449***	5.188***	5.396***
	(1.474)	(1.454)	(1.493)	(1.669)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Province-year FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
N	16,961	16,961	16,961	19,496
R^2	0.824	0.845	0.872	0.877

Outcome is county pharmacy count. Controls include log population, log nighttime lights, and log road density unless otherwise noted.

TWFE DID: Log Count and Extensive Margin

	ln FE	Has FE	ln prov-year	Has prov-year
Post policy	0.020	0.030***	0.026	0.025***
	(0.014)	(0.004)	(0.016)	(0.005)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Province-year FE	No	No	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	15,469	16,961	15,464	16,961
R^2	0.882	0.664	0.906	0.740

Log count excludes zero-count counties. Extensive margin is an indicator for having at least one pharmacy.

Continuous DID by Baseline Public-hospital Intensity

	Count FE	ln Count FE	Count prov-year	ln Count prov-year
Post policy $\times$ ln(public hospitals in 2012 + 1)	20.672***	-0.068***	12.179***	-0.118***
	(3.776)	(0.024)	(3.392)	(0.024)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Province-year FE	No	No	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	16,961	15,469	16,961	15,464
R^2	0.847	0.882	0.873	0.906

Counties with more baseline public hospitals show larger post-reform level-count increases, but lower log-count estimates.

CSDID Average Effects: Count and Log Count

	Pharmacy count	Log pharmacy count
ATT	-2.178	0.010
	(2.077)	(0.032)
Pre-average	-2.403	0.038
	(2.696)	(0.077)
Post-average	-4.021	-0.008
	(2.413)	(0.039)
Controls	Yes	Yes
Control group	Not-yet-treated	Not-yet-treated

CSDID estimates are from Callaway and Sant'Anna style not-yet-treated comparisons. Event windows are shown only where later periods have control support.

PPML Count Regressions

	FE	Prov-year
Post policy	-0.014	-0.014
	(0.012)	(0.015)
County FE	Yes	Yes
Year FE	Yes	No
Province-year FE	No	Yes
Controls	Yes	Yes
N	16,947	16,929

PPML keeps zero-count observations. Unlike TWFE level-count estimates, PPML does not show a statistically significant count response.

Chain vs Non-chain Outcomes

	Chain count	Non-chain count	Chain share
Post policy	3.578***	1.609	0.014***
	(0.354)	(1.357)	(0.002)
County FE	Yes	Yes	Yes
Province-year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
N	16,961	16,961	16,961
R^2	0.724	0.858	0.747

The reform is associated with a larger increase in chain pharmacies and a higher chain share; the non-chain count effect is positive but imprecise.

Pharmacy Distance to Nearest Hospital

	Mean km FE	Mean km prov-year	Share ≤2km FE	Share ≤2km prov-year
Post policy	0.000	0.100	0.001	0.001
	(0.124)	(0.155)	(0.005)	(0.006)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
Province-year FE	No	Yes	No	Yes
Controls	Yes	Yes	Yes	Yes
N	15,469	15,464	15,469	15,464
R^2	0.666	0.677	0.740	0.755

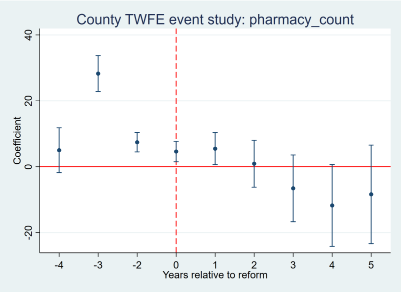
Outcome variables are county-year averages or shares based on each pharmacy's distance to its nearest hospital.

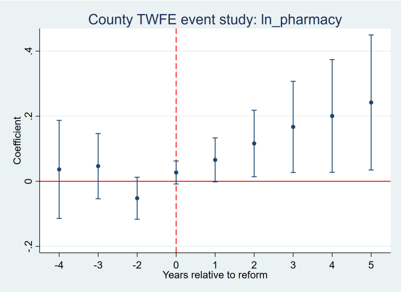
Pharmacy Counts by Distance to Nearest Hospital

	$\leq$2km FE	$>$2km FE	$\leq$2km prov-year	$>$2km prov-year
Post policy	5.314***	0.135	5.298***	-0.110
	(1.021)	(0.785)	(1.027)	(0.841)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	No
Province-year FE	No	No	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	16,961	16,961	16,961	16,961
R^2	0.849	0.763	0.867	0.803

The count increase is concentrated among pharmacies whose nearest hospital is within 2km; average distance and within-2km share are essentially unchanged.

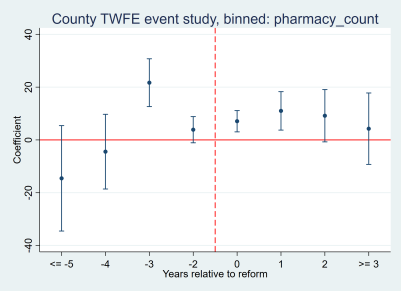
TWFE Event-study Diagnostics

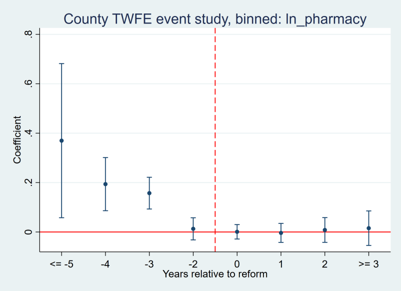




Standard TWFE event-study graphs; relative year -1 is omitted. Left: count. Right: log count.

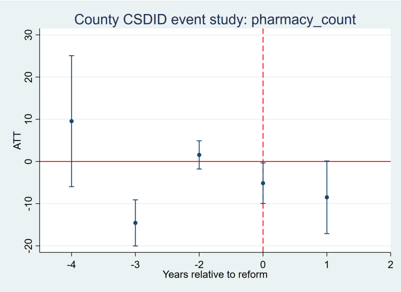
TWFE Event-study Diagnostics: Binned Version

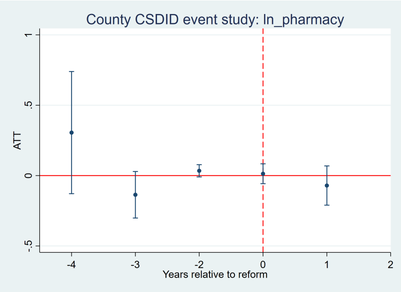




Binned version uses $F5/F4/F3/F2/L0/L1/L2/L3$, matching the five-estimator robustness graph.

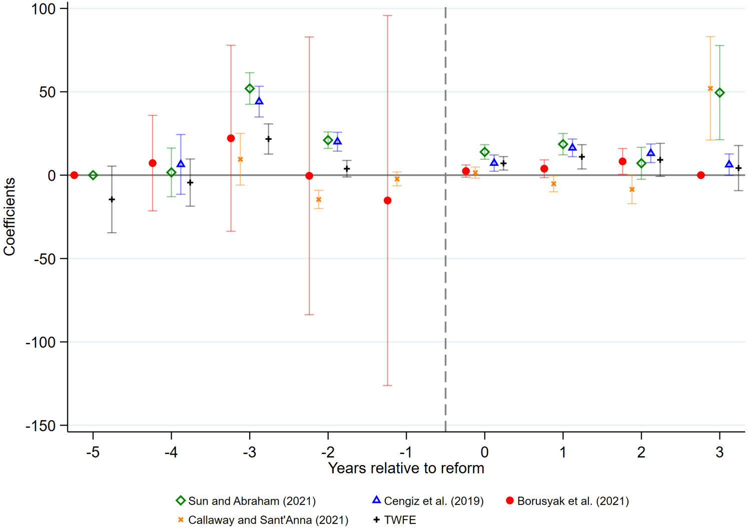
CSDID Event-study Diagnostics





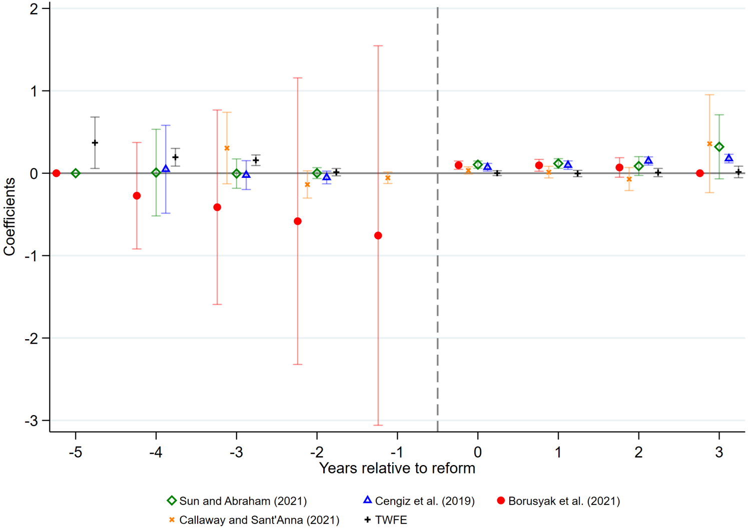
CSDID uses not-yet-treated counties as controls. Later post-periods without control support are not plotted.

Five-estimator Event-study Robustness: Count



Comparison of Sun-Abraham, Cengiz stacked event study, Borusyak-Jaravel-Spiess imputation, Callaway-Sant'Anna, and TWFE.

Five-estimator Event-study Robustness: Log Count



Same five-estimator comparison for log pharmacy count.

Summary of County-level Results

- **TWFE count regressions are positive and stable.** The preferred province-year specification implies roughly 5.2 additional pharmacies after reform.
- **Log count estimates are small, while the extensive margin increases.** Log-count coefficients are imprecise; the probability of having any pharmacy rises by 2.5–3.0 percentage points.
- **Continuous DID suggests heterogeneity by baseline public-hospital intensity.** Counties with more public hospitals have larger level-count increases, but lower log-count coefficients.
- **CSDID and PPML are more muted.** CSDID average ATT and PPML count regressions do not show statistically significant average effects.
- **Distance results do not support a broad relocation away from hospitals.** Average distance and within-2km share barely change; count increases are concentrated within 2km.
- **Composition shifts toward chains.** Chain pharmacy counts and chain share increase significantly; non-chain count changes are imprecise.